

No. of Functional Features Included in the Solution

Date	7 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Authentication	Secure user Login and Sign-up system using encrypted passwords.	Security / Auth	Done	
2	Study Field Context	A dropdown allowing students to specify their field of study for context-aware AI advice.	User Interface	Done	
3	Emotion Prediction Engine	Evaluates student text through deep learning models (BiLSTM & BERT) to detect emotions like frustration or anxiety.	Backend / Logic	Done	
4	Empathy Response Generator	Passes the predicted emotion to the Gemini API to formulate a personalized academic support strategy.	API / Integration	Done	
5	Quick Examples	Interactive buttons that instantly auto-fill the text area to allow for rapid testing of the application.	User Interface	Done	
6	History Logger	Automatically saves every interaction (emotion scores, AI response) into an SQLite database and local CSV.	Database / Storage	Done	
7	Analytics Dashboard	Dynamically generates a Pie Chart visualizing the distribution of the student's emotions over time.	User Interface	Done	
8	Emotional Timeline	Dynamically generates a Line Chart plotting the student's	User Interface	Done	

		confidence levels and emotional journey.			
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Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (<i>Auth, Emotion Engine, GenAI Response, DB Logging, Dashboard</i>)
Additional / Nice-to-Have Features	3 (<i>Quick Examples, Emotional Timeline Chart, Detailed JSON view</i>)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Dropdown menus, Quick Example buttons, Interactive Plotly Dashboards (Pie & Line charts).
2	Backend / Logic	1	NLP Text Preprocessing, BiLSTM / BERT Model Inference execution.
3	Database / Storage	1	SQLite user tracking, Pandas CSV historical interaction logging.
4	API / Integration	1	Google Gemini API prompt engineering and response fetching.
5	Security / Authentication	1	Secure Login & Signup handling using bcrypt password hashing.